

AT1 – Software Defined Datacenter (SDDC)

INTRODUCTION

A Software Defined Datacenter (SDDC) is a modern data center where compute, storage, networking, and security are controlled by software instead of manual hardware configuration. This makes the datacenter more flexible, faster, and easier to manage .

1. Traditional Datacenter

- Physical servers
- Manual configuration
- Slow deployment
- High maintenance cost

2. SDDC

- Everything managed through software
- Automation
- Fast provisioning
- Easy scaling

3. Three Virtualizations

Compute Virtualization

Storage Virtualization

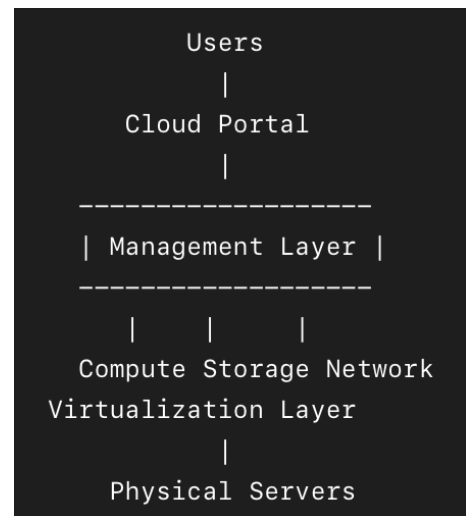
Network Virtualization

Advantages

- Faster deployment
- Lower cost
- Better resource utilization
- Automation
- High scalability

Conclusion

Compared to traditional datacenters, SDDC provides better agility, flexibility, automation, and efficient resource utilization.



AT2 – Multi-Region High Availability

Introduction

High Availability means an application should remain available even if one data center fails.

Multi-region Deployment

Application runs in two regions.

Example:

Chennai

Mumbai

If Chennai fails → Mumbai continues.

Data Replication

Data is copied continuously.

Benefits:

No data loss, Fast recovery

Failover

If one region crashes,

Load Balancer sends users automatically to another region.

Users don't notice downtime.

Monitoring

Examples:

AWS CloudWatch

Azure Monitor

Checks:

CPU

Memory

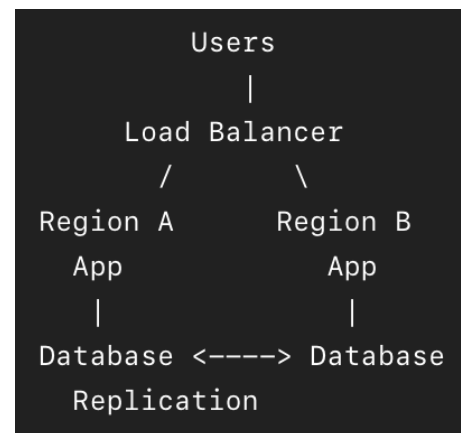
Disk

Network

Alerts administrators before failures.

Conclusion

Multi-region deployment improves availability, reliability, and customer satisfaction.



AT3 – Auto Scaling & Load Balancing

Introduction

During festivals like Amazon Great Indian Festival or Flipkart Big Billion Days, millions of users visit at the same time. Cloud platforms solve this using Auto Scaling and Load Balancing.

Load Balancer

Distributes traffic equally.

Without it:

One server gets overloaded.

With it:

Traffic is shared.

Auto Scaling

When traffic increases:

New servers are added.

When traffic decreases:

Extra servers are removed.

Monitoring

Tools:

AWS CloudWatch

Azure Monitor

Monitors:

CPU

RAM

Network

Triggers Auto Scaling.

Real Example

Amazon Great Indian Festival

Traffic:

Normal Day → 5 servers

Sale Day:

Auto Scaling increases to 100 servers

After Sale:

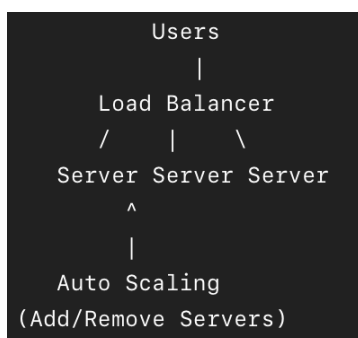
Returns to 5 servers

Customers experience:

Fast website

No crashes

Better shopping experience



Conclusion

"Auto Scaling and Load Balancing ensure cloud applications remain fast, reliable, and cost-effective even during sudden traffic spikes."

